

Weather AI Analytics

A Localized Weather Forecasting Solution

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- **Global Context:** Demand for precise, localized weather information.
- **The Problem:** Domination of resource-heavy models.
- **Objective:** Lightweight, computationally efficient solution.

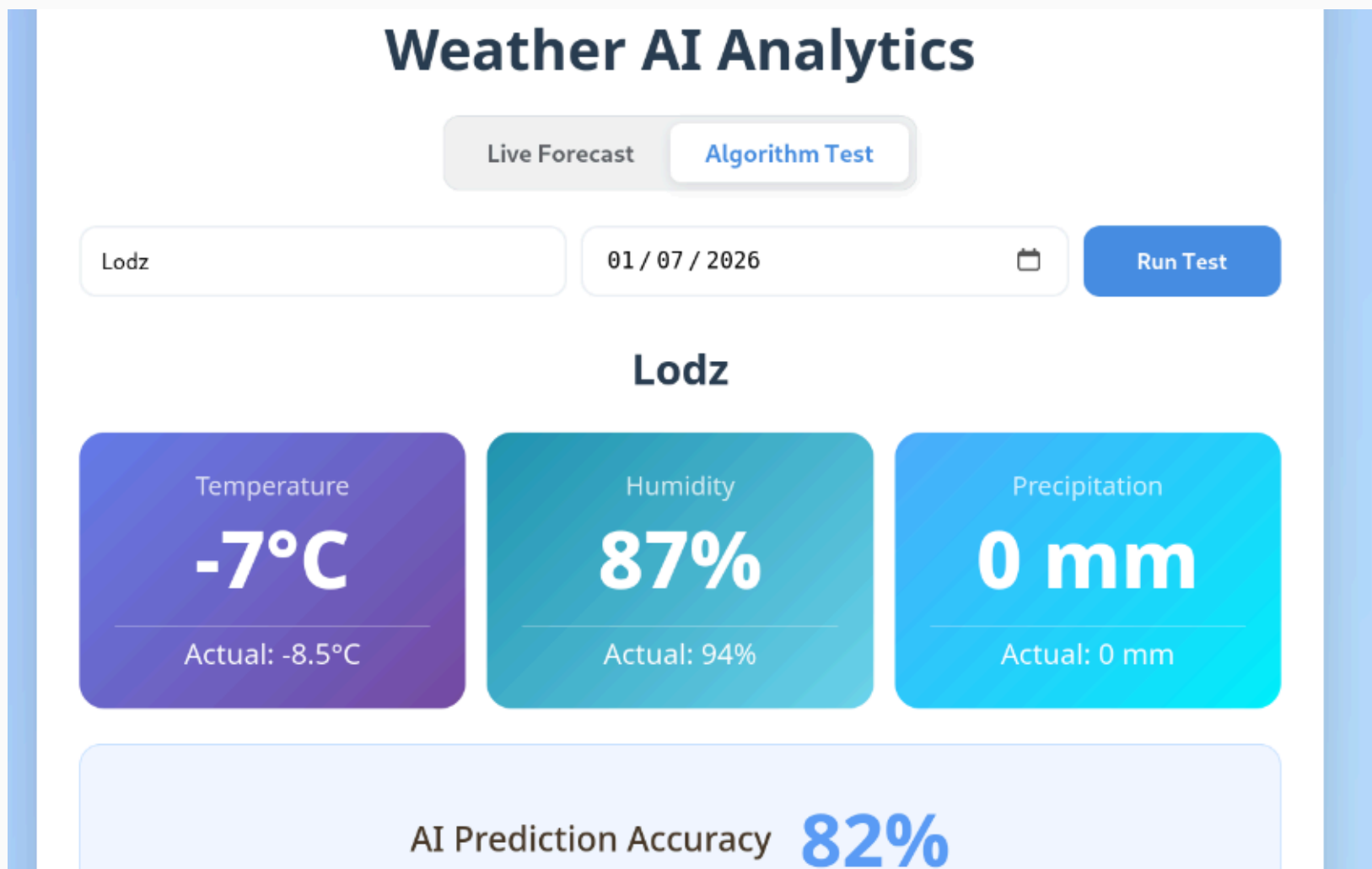
Two scientific pillars:

- **Persistence and Momentum Theory:** Atmosphere as a fluid system.
- **Chaotic Volatility:** Sensitive to immediate localized fluctuations.

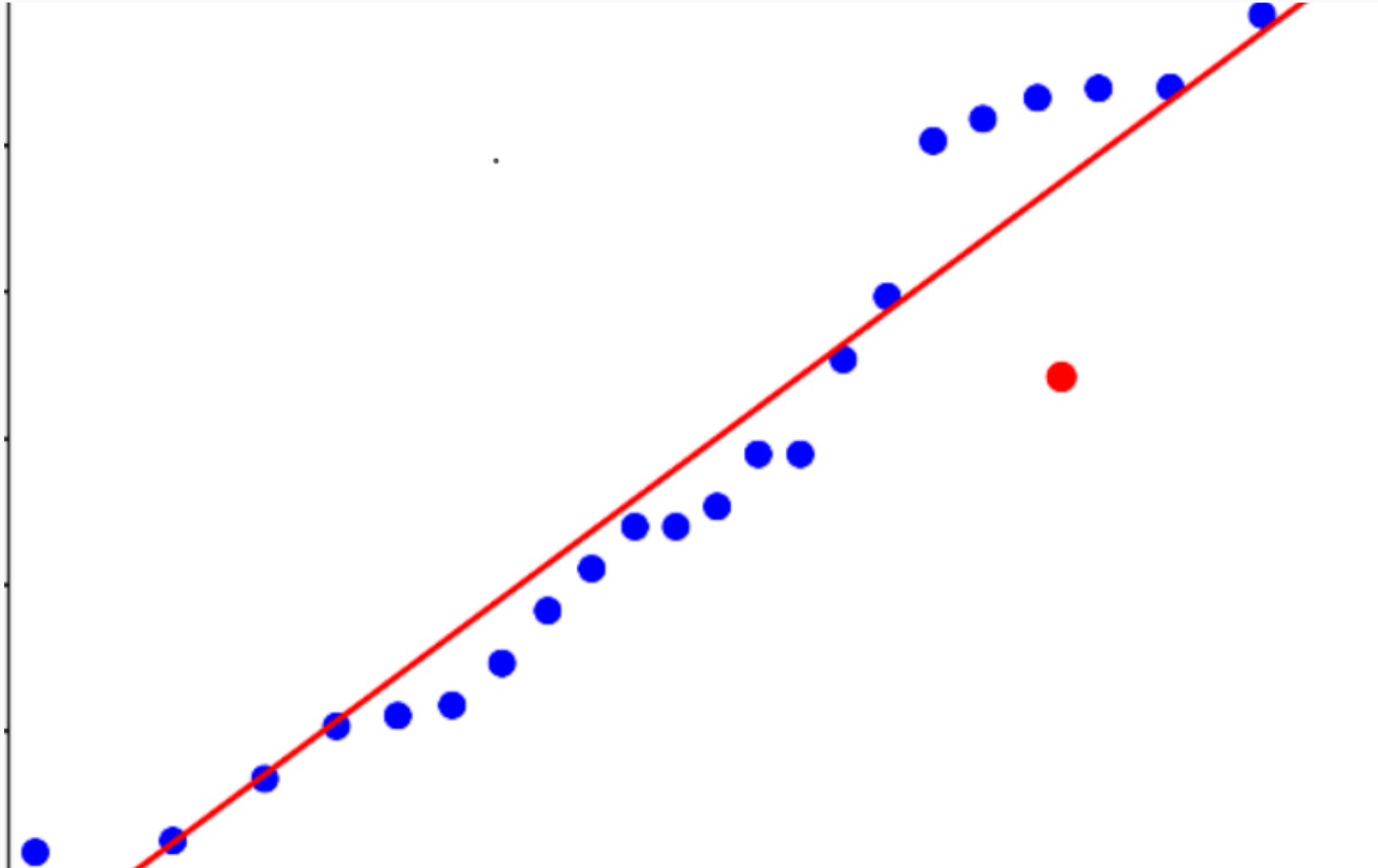
- **The goal:** Prove the correctness of algorithm.



- **Backend:** Flask.
- **Frontend:** SPA with dynamic elements.
- **Data Orchestration:** Open-Meteo API for historical training sets.



Challenges: Dealing with Outliers



Challenges: Dealing with Outliers

- **Portability Solution:** Deployment on low-power 8-bit microcomputers.
- **Edge Intelligence:** Local forecasting without Internet.



- **Summary:** Verifiable, low-latency alternative.
- **Future Work:** Thermometer prototypes.
- **Applications:** Technology utilizing in developing countries.

- **Project:** <https://github.com/DvorNil/Weather-forecasting/>
- **Open-Meteo API:** <https://open-meteo.com/>
- **Linear Regression:** Montgomery, D. C. et al. (2021)
- **EWMA Foundation:** Hunter, J. S. (1986). “The Exponentially Weighted Moving Average”